

Research Article

MINN: A metabolic-informed neural network for integrating omics data into genome-scale metabolic modeling

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ABSTRACT

The understanding of cellular behavior relies on the integration of metabolism and its regulation. Multi-omics data provide a detailed snapshot of the molecular processes underpinning cellular functions and their regulation, describing the current state of the cell. While Machine Learning (ML) models can uncover complex patterns and relationships within these data, they require large datasets for training and often lack interpretability. On the other hand, mathematical models, such as Genome-Scale Metabolic Models (GEMs), offer a structured framework for analyzing the organization and dynamics of specific cellular mechanisms. At the same time, they don't allow for seamless integration of omics information. Recently, a new framework to embed GEMs in a neural network has been introduced: these hybrid models combine the strengths of mechanistic and data-driven approaches, offering a promising platform for integrating different data sources with mechanistic knowledge. In this study, we present a Metabolic-Informed Neural Network (MINN) that utilizes multi-omics data to predict metabolic fluxes in *Escherichia coli*, under different growth rates and gene knockouts. We test its performances against pure ML and parsimonious Flux Balance Analysis (pFBA), demonstrating its efficacy in improving prediction performances. We also highlight how conflicts can emerge between the data-driven and the mechanistic objectives, and we propose different solutions to mitigate them. Finally, we illustrate a strategy to couple the MINN with pFBA, enhancing the interpretability of the solution.

1. Introduction

The phenotype of a cell is a complex interplay between its metabolic network, consisting of thousands of biochemical reactions, and the regulatory mechanisms controlling diverse cellular functions. Mechanistic models, such as GEMs [39], provide a structured framework to integrate and connect the available knowledge to find emergent properties in cellular systems. GEMs mathematically represent cellular metabolism, summarizing our information about the biochemical processes present in an organism [28,34,37]. One common approach to simulate cellular behavior using GEMs is constraint-based modeling (CBM). Among these methods, Flux Balance Analysis (FBA) [33,8,35] is particularly notable. FBA applies linear programming to optimize the distribution of metabolic fluxes, aiming to maximize specific objectives like biomass

production while considering nutrient availability constraints. However, the predictive power of a mechanistic model like FBA is limited by the completeness of our understanding of cellular processes. Moreover, FBA typically has multiple feasible solutions. In such cases, the solution with the lowest sum of fluxes is usually selected, based on the assumption that cells try to minimize their enzyme production [23]. However, this assumption is often an oversimplification, which does not account for the complex regulatory mechanisms within cells.

Omics data can be integrated in GEMs to enhance their predictive power and tailor models to specific cellular contexts [26,40]. Transcriptomics and proteomics offer indirect and direct proxies for metabolic activity, respectively, and have been incorporated into GEMs using tools such as GIMME [5] and CoCo [41] for gene expression data, and GECKO [9] and sEnz [6] for enzyme abundances. Metabolomics and fluxomics

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provide additional constraints through extracellular metabolite levels or isotope-labeling experiments [17,4]. While multi-omics integration offers a more comprehensive view of the cellular state [25], these efforts remain limited by standardization challenges and the difficulty of mechanistically linking non-metabolic features to model reactions [24].

On the other hand, data-driven machine learning (ML) models can effectively extract patterns in high-dimensional datasets, such as multi-omics data, without prior knowledge of underlying molecular mechanisms. These models often demonstrate strong predictive capabilities, but are limited by the scarcity of biological datasets, frequently constrained by experimental costs and time. In recent years, ML models have also been explored for predicting metabolic fluxes using multi-omics data. Although FBA remains the preferred mechanistic framework for this task, integrating omics data into CBM frameworks remains a significant challenge [26]. Interestingly, recent work [16] demonstrated that purely ML-based approaches trained on omics data can outperform FBA-based methods in metabolic flux prediction. A more detailed overview of the literature is provided in Section 1 of the Supplementary Materials.

Given the complementary strengths and limitations of ML and GEMs, there has been increasing interest in trying to combine these two approaches [42,3,21]. Hybrid models that merge mechanistic knowledge with the predictive capabilities of ML offer a promising direction but so far, as highlighted in Sahu et al. [36], the existing applications do not truly integrate ML and FBA. Instead, they mostly combine them in two separate steps: using ML as input for FBA [12,20,30], or using FBA as input for ML [27,10].

Recently, Faure et al. [13] developed a framework that truly combines CBM with ML in a neural network architecture called an Artificial Metabolic Network (AMN). Their approach leveraged GEM structures and FBA constraints within neural networks to predict growth rates from media compositions.

In Faure et al. [13], the authors suggested three different possible configurations for incorporating the GEM structure and the FBA constraints in a neural network (NN). In this work, we selected one of these configurations, inspired by Physics-Informed Neural Networks (PINNs) [11], and we re-implemented and expanded it to integrate multi-omics data as inputs. We will refer to this new architecture as Metabolic-Informed Neural Network (MINN) with multi-omics integration (Fig. 1). We applied this hybrid model to the dataset analyzed by Gonçalves et al. [16], which examines how the metabolism of *Escherichia coli* adapts to varying growth rates and single-gene knockouts [19]. As discussed in Tazza et al. [38], the combination of fluxes measured experimentally lies outside the solution space of FBA, causing a conflict between the optimization of the data-driven and the mechanistic objectives. To address this, we provide different mitigation strategies. To summarize, in this work:

- i. We describe the implementation of a MINN with multi-omics integration, using an early concatenation approach.
- ii. We benchmark its predictive performances on the ISHII dataset, compared to pure ML methods [16].
- iii. We recalculated the data to be in the FBA solution space and compared the predictive performances with those based on the original measurements.
- iv. We explore different hybrid optimization strategies to address the conflict between the objectives, while using the original data.
- v. Finally, we adapt the MINN to a “reservoir” configuration Faure et al. [13], which uses the MINN predictions to directly constrain pFBA, and compare its predictions with those of pFBA alone.

With these analyses, we provide a detailed overview of methods and strategies to adapt and use hybrid ML-FBA methods for multi-omics integration. Our findings highlight the potential of hybrid models to enhance the predictive accuracy and robustness of metabolic flux predictions. This paves the way for more precise and comprehensive metabolic

network analyses, particularly for phenotypes where metabolism is significantly influenced by other layers of cellular organization, which are challenging to incorporate into FBA. Furthermore, with this work we aim to provide a guide to the use of the MINN framework, helping researchers choose the most suitable configuration based on the specific objective of their study.

2. Methods

2.1. Dataset

The dataset analyzed in this work was originally published by Ishii et al. [19] and consists of 29 chemostat experiments, in which *E. coli* was grown in glucose minimal medium. Wild-type strain K-12 was grown at 5 different dilution rates ($D = 0.1, 0.2, 0.4, 0.5$, and 0.7 h^{-1}), while 24 different single-knockout mutant strains were cultivated at fixed dilution rate ($D = 0.2 \text{ h}^{-1}$). The same dataset was already used by Gonçalves et al. [16] to test traditional ML for the prediction of metabolic fluxes from multi-omics data.

The dataset consists of transcriptomic, proteomic, and fluxomic measurements. For each sample, microarrays were used to assess the expression profiles of 79 genes and LC-MS/MS quantitative proteomics to measure the abundances of 60 proteins. ^{13}C -labeled metabolomics experiments were analyzed with MFA to estimate 47 metabolic fluxes: 37 reactions of the central carbon metabolism, 9 exchange fluxes (production or consumption of external metabolites) and biomass growth.

The metabolic model used by Ishii et al. [19] to perform MFA is a core model that mainly represents the central carbon metabolism of *E. coli* and how it connects to the measured external metabolites. This model is much smaller and less complete than the GEM [14] integrated in the MINN. For the GEM to grow, many more different biomass components must be synthesized, diverting some metabolic precursors outside the pathways represented in the MFA model. For this reason, the fluxomics data from Ishii et al. [19] lie outside the solution space [38]. In most of our analyses we used the original fluxomics data, to highlight the ability of the MINN to reconcile MFA fluxomics data with the structure of the full-size GEMs. However, to investigate the impact of this discrepancy, we repeated some of the analyses with a second set of fluxes, now residing in the FBA solution space. This second set of fluxomics data is composed of the fluxes with the minimum Euclidean distance from the original ones, following an approach detailed in the Supplementary Material of Machado and Herrgard [26] and we refer to it as *FBA fit* data.

2.2. GEM preparation

In this section we describe all the genome-scale metabolic reconstructions utilized to build the MINNs. The most recent GEM available for *E. coli* K-12 is iML1515 [29], but we opted for iAF1260 [14]. The two differ mainly for the more comprehensive coverage of accessory pathways of iML1515, which are relevant in complex environments like the human gut, but not for growth on minimal medium. On the other hand, the size of the GEM can heavily affect the complexity of the MINN: using a smaller GEM would improve the efficiency of our hybrid model by reducing the computational resources required for training. Possibly, it would also reduce the noise in the model, enhancing the prediction accuracy. iAF1260 is reasonably smaller than iML1515 (2382 reactions vs. 2712) and is also the same model used by Gonçalves et al. [16] in their analyses.

We further reduced the size of the model by excluding all the reactions which cannot carry flux during growth in glucose minimal medium. This was achieved performing Flux Variability Analysis (FVA) and retaining only the reactions with a non-zero span. We refer to this model as *FVA-reduced* GEM. We also tested a second strategy, inspired from Faure et al. [13], to further reduce the model. We generated a

Table 1

Dimensions of all the GEM used in this analysis.

GEMs		
GEM name	original splitted reactions	reduced and splitted reactions
iAF1260	2957	NA
iAF1260 FVA-reduced	2957	1873
iAF1260 FBA-reduced	2957	587
e_coli_core	115	NA
iNF517 FVA-reduced	1022	704

dataset of 2000 FBA solutions by randomly selecting single-gene knock-outs and varying the maximum glucose uptake rate within the experimentally observed range. Reactions that consistently carried zero flux across all the simulations were removed from the model. We refer to this model as *FBA-reduced* GEM. To investigate the impact of an extreme decrease in the genome-scale reconstruction size, we also built a MINN using the *e_coli_core* model [32], a manually reduced GEM focused on central carbon metabolism, which is the smallest model available in the BiGG database.

Finally, to further test the role of the GEM and the underlining metabolic network, we also tested our baseline configuration including the GEM of a different organism. We used the iNF517 model [15] for *Lactococcus lactis subsp. cremoris* MG1363. This microorganism is a lactic acid bacterium, with an incomplete TCA cycle, which makes it an interesting comparison for *E.coli*, both in terms of structure of the network in the central carbon metabolism and of general metabolic behavior. The model was reduced using the FVA-guided reduction approach. The results of this comparison are available in the Section 3.1 of the Supplementary Materials.

In Table 1 we summarize the dimensions of each GEM. The GEMs were downloaded from the BiGG database [22] and handled/modified using CBMPy 0.8.4 [31]. In each model, reversible reactions were split into a forward and a reverse reaction using the built-in CBMPy function `cbmpy.CBTools.splitReversibleReactions`.

2.3. MINN architecture

This study introduces a MINN architecture designed to predict multiple fluxes using multi-omics data, which provide key insights for metabolic predictions but are challenging to integrate with FBA [26]. Fig. 1 illustrates the structure of our MINN architecture, built to predict fluxes measured in the ISHII dataset using proteomics, transcriptomics, and the measurements of two exchange fluxes, namely $R_{EX_glc_D_e}$, $R_{EX_o2_e}$. The data are integrated using an early concatenation strategy [1], where the three omics datasets are combined into a single matrix that is fed into the MINN.

The omics data are used in the first part of the model (shown in red in Fig. 1), which is a pure feed-forward neural network. Here, the network is trained to learn a mapping from the input omics profiles to an initial estimate of the flux distribution, denoted as V_0 . The input dimension d_{in} corresponds to the total number of features after concatenating the transcriptomics, proteomics, and exchange flux data, while the output dimension d_{out} corresponds to the total number of reactions in the GEM. This part of the model is purely data-driven, meaning that it does not rely on any mechanistic assumption or require prior knowledge such as gene-protein-reaction (GPR) associations. Instead, it learns directly from the data how to associate the omics features with a plausible flux configuration. This makes the method flexible and compatible with a wide range of input omics data, including those not directly related to metabolism but still informative of the broader cellular context.

The second part, the mechanistic layer, blue in Fig. 1, consists of a gradient descent optimization loop. This loop refines the output of the first neural network step by adjusting the final predicted flux distribution V to better comply with the FBA constraints, minimizing the FBA loss function L_{FBA} . The neural network weights are trained using a

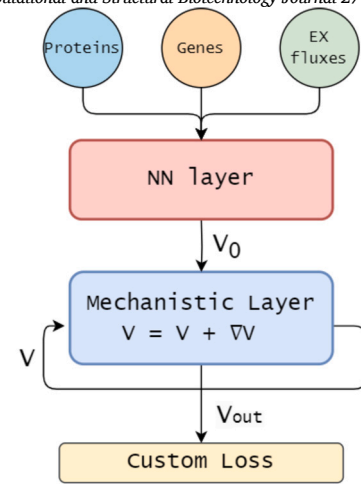


Fig. 1. Schematic representation of the MINN architecture. Protein and gene expression levels, and exchange flux data are used as input to a feed-forward neural network, which produces an initial estimate for the flux distribution V_0 . This estimate is refined in a mechanistic layer via a gradient descent step to better align with flux balance constraints, resulting in the final flux distribution V_{out} . The custom loss function combines the discrepancy between the model predictions and the target fluxomics data with the violation of FBA constraints, and is used to train the network via backpropagation.

standard back-propagation algorithm and a custom loss function L_{MINN} that considers both the data error and the FBA constraints.

To formalize this, let the input data be $X \in \mathbb{R}^{N \times d_{in}}$ where N is the mini-batch dimension in the back-propagation algorithm and d_{in} is the number of features of X . The first NN part can be expressed as:

$$V_0 = \sigma(XW^h + b^h)W^{out} + b^{out} \quad (1)$$

with σ the ReLU activation function, $W^h \in \mathbb{R}^{d_{in} \times d_h}$, $b^h \in \mathbb{R}^{1 \times d_h}$, $W^{out} \in \mathbb{R}^{d_h \times d_{out}}$, $b^{out} \in \mathbb{R}^{1 \times d_{out}}$, the weight matrices and the biases of the input and hidden layer respectively, where d_h is the hidden layer dimension and d_{out} is the output dimension, which coincides with the dimension of the flux distribution. Then, V_0 is refined in the mechanistic layer through a gradient descent optimization, using only the mechanistic constraints. For simplicity, the notation refers to the simple case when the loop has one iteration:

$$V_{out} = V - lr \frac{\partial L_{FBA}}{\partial V} \quad (2)$$

$$L_{FBA} = \frac{1}{m} |SV|^2 + \frac{1}{n_{in}} |\text{ReLU}(P_{in}V - V_{in})|^2 + \frac{1}{n} |\text{ReLU}(-V)|^2$$

The first element represents the steady-state constraint of FBA, with S as the stoichiometric matrix of the GEM. The term m denotes the number of metabolites and serves as the normalization term. The second element represents the upper-bound constraint on the vector of fluxes V_{in} . Here, P_{in} represents the projection matrix that projects the flux distribution vector V into the dimension of V_{in} , while the normalization term n_{in} stands for the number of bounded fluxes. Lastly, the last element symbolizes the lower bound constraint, which required since the GEM is built to ensure that all the fluxes are positive.

The custom loss used to train the weights of the MINN is:

$$L_{MINN} = L_1 + L_2 + L_3 + L_4 \quad (3)$$

$$= \frac{|P_{ref}V - V_{ref}|}{V_{ref}} + \frac{1}{m} |SV|^2 + \frac{1}{n_{in}} |\text{ReLU}(P_{in}V - V_{in})|^2 + \frac{1}{n} |\text{ReLU}(-V)|^2$$

where V_{ref} is the vector with the measured fluxes and P_{ref} a projection matrix that projects V to the dimension of V_{ref} ; while the other elements represent the FBA constraints and are the same as in L_{FBA} . In order to guide the reader through the understanding of the MINN

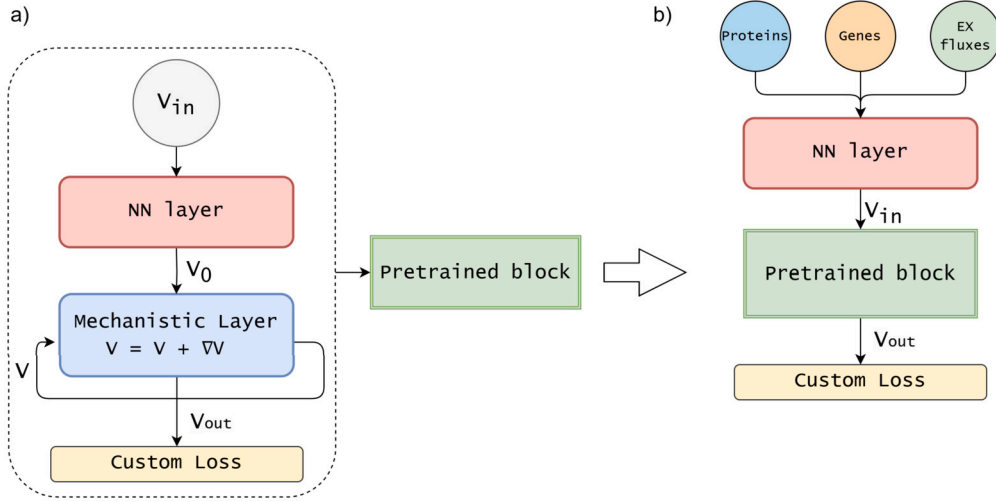


Fig. 2. Two-step training strategy of the MINN-reservoir architecture: a) In the first step, a MINN (with no omics data in input) is trained to approximate an FBA solver, using a dataset of simulated FBA solutions. The network learns to predict the flux distribution V_{out} from randomly sampled external fluxes V_{in} ($R_{EX_glc_D_e}$, $R_{EX_o2_e}$, $R_{EX_co2_e}$, $R_{EX_etoh_e}$ and $R_{EX_ac_e}$). Once trained, its weights are frozen, and the resulting model is reused as a fixed *Pretrained block*. b) In the second step, this *Pretrained block* is embedded within a new architecture that takes omics data and medium exchange fluxes ($R_{EX_glc_D_e}$, $R_{EX_o2_e}$) as input. A neural network predicts V_{in} , which is then passed to the *Pretrained block* to compute V_{out} .

architecture, we provide an illustrative toy example in Section 1 of the Supplementary Material.

In Faure et al. [13], the authors used Mean Squared Error (MSE) as L_1 because they wanted to predict a single flux, specifically the growth rate. In our work, we use the Normalized Error (NE) [16] to have a scale-invariant L_1 when predicting multiple fluxes in order to avoid favoring reactions with higher flux values.

In addition, during our analysis a conflict between the data-driven and mechanistic losses emerged. In order to mitigate this issue, we multiply L_1 with a constant c , which allow us to adjust the balance between the two losses:

$$L_{\text{MINN-balanced}} = c \cdot L_1 + L_2 + L_3 + L_4 \quad (4)$$

The c constant becomes a hyperparameter of the model, tuned using k-fold cross-validation and the optimized value determines the best balance between L_1 and ($L_2 + L_3 + L_4$).

It is important to note that the mechanistic part of the loss function includes only terms enforcing FBA constraints, to reduce the solution space, but does not include any term related to the FBA objective (e.g., biomass maximization). As a result, the optimization is guided solely by L_1 , a data-driven objective, without imposing any predefined metabolic goal. This is particularly advantageous in cases where no clear cellular objective exists, such as in gene knockout mutants.

2.3.1. MINN-reservoir

Similarly to Faure et al. [13], we tested an additional configuration of the MINN, named MINN-reservoir. The training of the MINN-reservoir requires two steps, as shown in Fig. 2. In the first step (2a), a MINN (with no omics data in input) is trained only to reproduce FBA using a dataset of FBA solutions. This dataset contains the results of 2000 FBA simulations, in which the reactions belonging to V_{in} ($R_{EX_glc_D_e}$, $R_{EX_o2_e}$, $R_{EX_co2_e}$, $R_{EX_etoh_e}$ and $R_{EX_ac_e}$) were assigned random values, within the ranges of variability observed in the ISHII dataset. This procedure creates a model that acts as a pure approximator of an FBA solver (Pretrained block in Fig. 2), capable of predicting the optimal flux distributions from measurements of external metabolite fluxes, similar to an FBA solver. In the second step (Fig. 2b), this Pretrained block is embedded into a new MINN architecture, where it replaces the Mechanistic Layer. The resulting architecture consists of a neural network layer that predicts V_{in} from multi-omics data and medium exchange fluxes ($R_{EX_glc_D_e}$, $R_{EX_o2_e}$), followed by the *Pretrained block*, which

computes the flux distribution V_{out} from the predicted V_{in} . The two-step approach ensures that the predicted V_{in} values are compatible with FBA and can be reliably used by the solver to produce a true, linear programming solution. For the test sample in each split, the predicted V_{in} are extracted and used as additional constraints for pFBA: increasing the input information in a data-driven way while preserving the mechanistic structure of the model. Unlike the default configuration of the MINN, this approach produces as final output not only the full flux distribution, but a complete solution from a Linear Programming solver, which can be analyzed with all the tools developed for this purpose.

2.3.2. Hybrid optimization strategies for data-driven and mechanistic integration

Equation (3) highlights how the loss of the MINN is composed of two components: L_1 , which drives the optimization on the data, and $L_{FBA} = L_2 + L_3 + L_4$, which minimizes the divergence from the mechanistic constraints. In Equation (4), we already introduced the coefficient c , which allows us to tweak the balance between the two components, either manually or through hyperparameter optimization. In this section, we introduce three other methods to tune this balance while minimizing the trade-off between the two components.

In developing hybrid models that integrate mechanistic constraints with data-driven approaches, we propose different strategies to balance the objectives of maintaining adherence to the FBA constraints without substantially compromising predictive performance. These methods address the challenge posed by different scales of mechanistic and data-driven losses, ensuring that neither dominates the optimization process and that the model generalizes well to unseen data.

Bound on mechanistic loss The first method introduced is a bound on the mechanistic loss. This method ensures that the model's predictions do not deviate too far from the mechanistic solutions. A fixed threshold is set, and when the mechanistic loss exceeds this threshold, a multiplicative factor is applied to penalize further deviations. This approach softly constrains the model within a feasible solution space derived from mechanistic constraints, preventing the model from paying an excessive cost in terms of mechanistic loss to improve the data-driven one. To provide a clear view of the described bound on the mechanistic loss, a graphical representation is available in Section 2 of the Supplementary Material file.

Table 2

Comparison of predictive performance between our proposed MINN-based approaches and purely mechanistic and machine learning methods from Gonçalves et al. [16]. Metrics average and standard deviation over 29 leave-one-out splits. All the MINN models were generated using the iAF1260-FVA reduced GEM.

Model	ISHII			
	R^2	MAE	RMSE	NE
pFBA*	0.823 ± 0.156	0.692 ± 0.733	1.058 ± 1.029	0.381 ± 0.185
NN*	0.967 ± 0.036	0.652 ± 0.945	0.936 ± 1.314	0.338 ± 0.338
RF*	0.970 ± 0.037	0.507 ± 0.804	0.729 ± 1.105	0.271 ± 0.347
MINN-MSE-base	0.950 ± 0.060	0.525 ± 0.525	0.736 ± 0.703	0.287 ± 0.282
MINN-unbalanced	0.951 ± 0.051	0.563 ± 0.739	0.814 ± 1.067	0.325 ± 0.442
MINN-c-balanced	0.950 ± 0.055	0.473 ± 0.480	0.678 ± 0.653	0.272 ± 0.280

*Results from Gonçalves et al. [16].

Loss balancing A loss balancing mechanism was employed to handle the different scales of mechanistic and data-driven losses. This method normalizes each loss dividing it by the exponential average of its previous values, as detailed in Hu et al. [18]. Through this approach, both losses are considered equally during gradient updates, preventing one from outweighing the other during the training process. The loss balancing ensures that the mechanistic loss, which would typically be underrepresented due to its smaller scale, contributes adequately to model optimization alongside the data-driven loss.

Loss weight scheduler Lastly, a dynamic loss weight scheduler was implemented to gradually shift the model's focus between the mechanistic and data-driven tasks over the course of training. For the first phase, the scheduler prioritizes the mechanistic loss, ensuring it starts from a solution closer to the mechanistic model's feasible space. As training progresses, there is a transition phase where the scheduler gradually increases the importance of the data-driven loss until it reaches the final phase, where the data-driven loss has a higher weight, guiding the model toward better predictive performance for the data-driven task. The transitions between the three training phases were defined based on the learning curves of the validation data from the inner K-Fold cross-validation loop (also used for hyperparameter optimization). The transition phase was triggered once the mechanistic loss had converged, which occurred at epoch 30. This phase lasted for 40 epochs, followed by a final phase of 80 epochs, during which the data-driven objective was given higher priority. The loss balance in the initial phase was fixed at 90% mechanistic and 10% data-driven. In contrast, the balance parameter for the final phase was subject to hyperparameter tuning, with a search space ranging from 80% to 100% data-driven loss (and the remainder assigned to the mechanistic loss). For the sake of clarity, a graphical representation of the weight scheduler is available in Section 2 of the Supplementary Material file.

These methods provide a comprehensive framework for balancing the trade-offs between data-driven accuracy and mechanistic integrity.

2.4. Computational setup

We adopted the same evaluation pipeline for all our MINN configurations to evaluate the MINN performance and have a fair comparison with the results obtained by [16]. It consists of a dual-loop cross-validation process. The outer loop is a leave-one-out, and in each train loop, there is an inner loop of a k-fold with $k = 5$ to tune the hyper-parameters. The tuning concerns the dimension of the first hidden layer, the learning rate of the NN, the intensity of dropout and L_2 regularization, and the c constant for the L_1 loss in the case of the MINN-c-balanced. Instead, for the MINN-scheduler model, only the hyperparameter controlling the balance between data-driven and mechanistic losses in the final training phase was tuned, where the data-driven component becomes predominant. The initial balance and the epoch marking the transition between phases were kept fixed. For a consistent comparison with the work of Gonçalves et al. [16], we employed identical metrics to evaluate all

our experiments: the regression coefficient R^2 , the mean absolute error (MAE), the root mean squared error (RMSE) and the normalized error (NE). As described later, in some of our results, we also report the L_2 as a metric to measure the quality of the predicted flux distribution.

3. Results and discussion

This work aims to compare the predictive performance of the MINN w.r.t. pure ML approaches and mechanistic models such as pFBA. For clarity, we divide all the results and discussions into three groups. The first one (Table 2) contains the results of the performance comparison between our approach and the pure ML ones presented in Gonçalves et al. [16]. Here, we evaluate the predictive performance of different methods on the 45 reference fluxes measured in the ISHII dataset. The second one (Table 3) includes the results of the comparison between different approaches employed to mitigate the issue of conflicting losses. Here, we compare the performance of the different methods on the measured fluxes and the quality of the predicted flux distribution, which we measure using, as a proxy, L_2 . The third group (Table 4), instead, compares the results obtained using the MINN-reservoir configuration with those of pFBA.

Lastly, we tested the impact of using different GEMs in the mechanistic layer of the MINN. In particular, when we included the GEM of *Lactococcus lactis* subsp. *cremoris*, a bacterium with an incomplete TCA cycle compared to *E. coli*, we observed a decrease in the quality of the predicted flux distribution. Although the difference was moderate, probably due to the conservation of central carbon metabolism, these results suggest that the biological relevance of the GEM has an important role in the regularizing effect of the mechanistic layer. A more detailed analysis is available in Section 3.1 of the Supplementary Materials.

3.1. MINN to predict measured fluxes

In the first group, as a baseline, we used the MINN architecture with an MSE as L_1 (MINN-MSE-base), as in Faure et al. [13]. As shown in Table 2, the results are already comparable with a Random Forest (RF), the best machine learning method in Gonçalves et al. [16], and better than the NN approach. To avoid potential bias from the large discrepancies (up to two orders in magnitude) between the values of the fluxes we are predicting, we replaced the MSE in L_1 with a NE. As detailed in the Section 2, we multiply L_1 with a constant c , optimized during the cross-validation. This method, incorporating the c parameter, is referred to as MINN-c-balanced, while the one without this adjustment as MINN-unbalanced. Although the change from MSE to NE ensures a scale-invariant L_1 , it also reduces its magnitude, amplifying the conflict between losses as the mechanistic constraints gain more influence. For this reason, the MINN-unbalanced obtained worse results w.r.t. the MINN-MSE-base, but the MINN-c-balanced shows the best results, achieving comparable or better performance than the RF in three out of four metric averages.

Table 3

Performance comparison of different methods addressing the issue of conflicting losses. Metrics average and standard deviation over 29 leave-one-out splits. All the models were generated using the iAF1260-FVA reduced GEM.

Model	ISHII				
	R^2	MAE	RMSE	NE	L_2
MINN-c-balanced	0.950 ± 0.055	0.473 ± 0.480	0.678 ± 0.653	0.272 ± 0.280	$8.75 \cdot 10^{-5} \pm 2.95 \cdot 10^{-4}$
MINN-c-balanced <i>FBA fit</i>	0.957 ± 0.061	0.489 ± 0.497	0.706 ± 0.720	0.295 ± 0.403	$2.98 \cdot 10^{-5} \pm 8.75 \cdot 10^{-5}$
MINN-bound	0.949 ± 0.050	0.548 ± 0.556	0.790 ± 0.779	0.308 ± 0.314	$7.1 \cdot 10^{-5} \pm 2.1 \cdot 10^{-4}$
MINN-scheduler	0.949 ± 0.058	0.581 ± 0.823	0.833 ± 1.146	0.299 ± 0.320	$1.60 \cdot 10^{-5} \pm 7.63 \cdot 10^{-5}$
MINN-scheduler-bound	0.946 ± 0.062	0.560 ± 0.551	0.806 ± 0.761	0.304 ± 0.287	$7.47 \cdot 10^{-5} \pm 3.78 \cdot 10^{-4}$
MINN-divided loss	0.951 ± 0.050	0.489 ± 0.471	0.703 ± 0.648	0.281 ± 0.289	0.22 ± 0.29

Moreover, the MINN-c-balanced shows a reduction in standard deviation. This suggests that the inclusion of biological constraints stabilizes the learning process, reduces overfitting, and leads to more robust and consistent predictions across the 29 leave-one-out splits. In addition, MINN not only learn flux values from data but also derive the entire flux distributions, as done by FBA simulations. These results suggest that MINN is a promising hybrid approach for flux prediction, showing consistent improvements in both average performance and standard deviation across all baselines, including pure mechanistic and ML methods. However, the statistical significance tests detailed in the Supplementary Material (Section 3.2) indicate that the difference between MINN-c-balanced and Random Forest is not statistically significant. This, together with the fact that our analysis was limited to a single microorganism, highlights the need for further investigations to better understand the effectiveness of hybrid approaches in the context of flux prediction. The challenge of predicting multiple fluxes, with significant variability in their values, was effectively addressed by substituting the MSE with an NE for the L_1 and adding the c constant that handled the emerging imbalances. The MINN's flexibility also makes it a powerful tool for integrating multi-omics and potentially other kinds of data with GEMs, enabling its application in diverse systems biology contexts.

3.2. MINN to predict a qualitative flux distribution

Regarding the second group of results, we aim to compare different methods to mitigate the issue of the conflicting losses already introduced in Section 2. We address this problem from two different points of view. First, we act on the mechanistic aspects of the MINN: the reference data. In contrast, the second perspective addresses the optimization process of the MINN, where we apply various hybrid optimization strategies discussed in Section 2.3.2. As a baseline, we employ our best method in terms of prediction performance, hence MINN-c-balanced. In the first case, we compare it with a configuration of the same MINN-c-balanced which uses different fluxes data, recalculated to be in the solution space of FBA. We described this process in Section 2.1. As shown in the first section of Table 3, this approach, named MINN-c-balanced *FBA fit*, does not reduce the quality of the fluxes prediction, represented by the four metrics, but it improves the L_2 by reducing its value by a third.

The results show that MINN maintains strong predictive performance even when the flux data are not in the FBA solution space. This suggests that its data-driven component can compensate for deviations from mechanistic constraints. However, using flux data that aligns with the FBA solution space improves the quality of the predicted flux distribution. This correction makes the optimization process easier by alleviating the issue of conflicting losses.

Regarding the hybrid optimization strategies, we compare the MINN-c-balanced with three other methods previously introduced in Section 2.3.2: *MINN-bound*, which penalizes violations of the mechanistic loss that exceed a threshold; *MINN-scheduler*, which gradually shifts the focus from mechanistic to data-driven loss during training; and *MINN-scheduler-bound*, which combines both approaches. From the second section of the table, we observe that the *MINN-c-balanced* model achieves

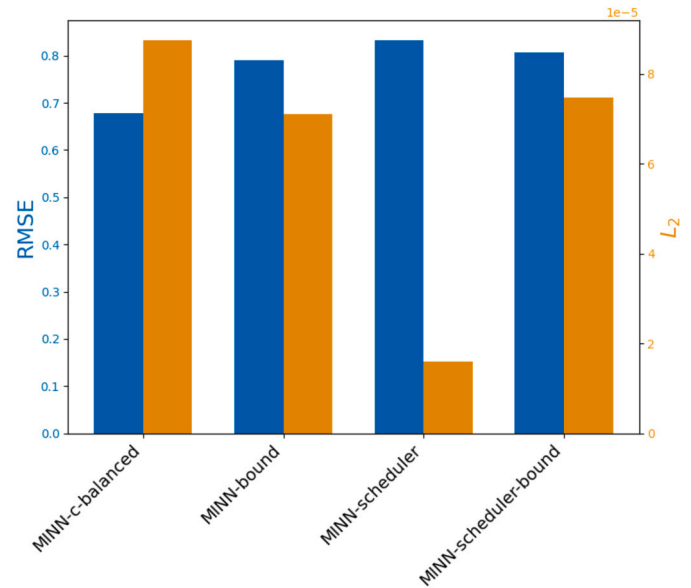


Fig. 3. Comparison of different methods based on data-driven task performance (RMSE) and mechanistic fit (L_2 loss), highlighting the trade-off between the two objectives.

the lowest RMSE, indicating the best performance on the data-driven task. However, this comes at the cost of a higher L_2 , suggesting a trade-off where improved performance is achieved at the expense of mechanistic fidelity. On the other hand, the models incorporating a mechanistic bound, such as *MINN-bound* and *MINN-scheduler-bound*, show a small increase in RMSE and a modest reduction in L_2 , suggesting a limited effect on improving the trade-off between mechanistic accuracy and data-driven performance. Interestingly, the *MINN-scheduler* model finds a better compromise between these objectives: at the cost of a moderate RMSE worsening, it achieves the lowest L_2 by a consistent margin. This demonstrates the strength of dynamic scheduling in keeping the solution close to the FBA feasible space, without heavily impacting data-driven performance. Interestingly, the balance parameter between the data-driven and mechanistic losses of the last phase, optimized during hyperparameter tuning, converged to values around 90–95% in favor of the data-driven loss, rather than the maximum of 100%, suggesting that the mechanistic component remained beneficial even during the data-driven-oriented phase.

Fig. 3 provides a visual comparison of the performance of these methods, focusing on the trade-offs between mechanistic fit and data-driven task performance. The *MINN-scheduler* model demonstrates a balanced performance across both objectives, with a moderate decline in data-driven accuracy but a substantial reduction in mechanistic loss, positioning it much closer to the FBA feasible solution. On the other hand, the models incorporating a mechanistic bound (*MINN-bound* and *MINN-scheduler-bound*) show improvements in mechanistic fit but at a comparable cost in terms of data-driven performance.

Table 4

Performance comparison between the standard pFBA and MINN-reservoir + pFBA. The results evaluate the effectiveness of the MINN-reservoir approach to enrich the input for pFBA in comparison to standard pFBA. Metrics average and standard deviation over 29 leave-one-out splits. The MINN-reservoir model was generated using the iAF1260-FBA reduced GEM.

Model	ISHII			
	R^2	MAE	RMSE	NE
pFBA	0.892 ± 0.127	0.496 ± 0.353	0.836 ± 0.625	0.306 ± 0.367
MINN-reservoir + pFBA	0.910 ± 0.091	0.445 ± 0.261	0.740 ± 0.421	0.253 ± 0.159

These results highlight a clear trade-off between optimizing for mechanistic fidelity and predictive accuracy. While the *MINN-c-balanced* method achieves the lowest RMSE, indicating better performance on the data-driven task, its high mechanistic loss shows that the model prioritizes predictive accuracy over adherence to mechanistic constraints. In contrast, the *MINN-scheduler* method effectively reduces the mechanistic loss, with only a marginal increase in RMSE.

Overall, these results emphasize the need to carefully select hybrid optimization methods based on the specific priorities of the task. In cases where mechanistic accuracy is crucial, dynamic scheduling methods like *MINN-scheduler* provide a balanced solution, allowing the model to gradually adjust the emphasis between mechanistic fidelity and data-driven optimization.

Additionally, as shown in the third section of Table 3, we tested a configuration of the MINN, called MINN-divided loss, where the loss is purely data-driven ($L_{MINN} = L_1$), to check if we could further improve the prediction performance at the cost of L_2 . The results show that the improvement in terms of prediction performance is marginal w.r.t. the substantial (four orders of magnitude) increase in the L_2 value. This confirms that the regularization effect happens exclusively in the Mechanistic Layer, while the complete L_{MINN} is needed to obtain a qualitative flux distribution as output of the MINN.

3.3. MINN-reservoir to improve pFBA predictions

In this third group, we present results for the MINN-reservoir method, which extends the use of MINN beyond applications focused solely on prediction. As introduced by Faure et al. [13], the MINN-reservoir can be used to generate constraints for a mechanistic model in a data-driven manner. In our case, we trained the MINN-reservoir to predict three arbitrarily selected exchange fluxes ($R_{EX_{co2_e}}$, $R_{EX_{etoh_e}}$ and $R_{EX_{ac_e}}$) which were then used as additional inputs for pFBA. FBA relies on optimization and is generally more accurate in predicting metabolic shifts caused by gene knockouts only after the microbial population has undergone an adaptation period. It is therefore of interest to explore whether the data-driven insights provided by the MINN-reservoir can help overcome this limitation in the case of the newly generated knockout strains from the ISHII dataset. Our baseline consists of a standard pFBA model that uses only the uptake rates of glucose ($R_{EX_{glc_D_e}}$) and oxygen ($R_{EX_{o2_e}}$) as inputs. We compare it with pFBA when it is provided with the same inputs plus the three extra constraints generated by the MINN-reservoir. This setup reflects a realistic use case, in which multi-omics data are available for all samples, while fluxomics data are only partially available. In this context, the MINN-reservoir allows us to estimate missing input fluxes from omics measurements, avoiding the need to experimentally quantify them for every sample.

The advantage of this neural network-based approach is that, once trained, it can generate additional inputs for pFBA based on initial conditions alone, enabling a more informative and automated modeling pipeline. As shown in Table 4, the enhanced version (MINN-reservoir + pFBA) slightly improves the performance in terms of average metrics across the 47 fluxes. However, the most evident benefit is the reduction

in the standard deviation, which indicates that the model produces more stable and consistent predictions across the 29 leave-one-out splits.

4. Conclusion

Faure et al. [13] introduced a new hybrid architecture, which incorporates a Genome-Scale Metabolic Model in a neural network structure, and used it to predict *E. coli* growth rates in different growth media. In our work, we adapted this framework into a Metabolic-Informed Neural Network, which also uses multi-omics data as input, and tested it with a more challenging task: predicting metabolic fluxes for different *E. coli* single-gene KO strains grown in minimal glucose medium. The MINN showed improved performance compared to traditional machine learning, and the mechanistic component exerted a regularizing effect on the predictions. We then explored the effect of different components of the architecture on the predictions and their accuracy. Finally, we tested the ability of the MINN to reconcile data and models in a flexible way, even in scenarios where data-driven and mechanistic optimization show a trade-off. To achieve this, we suggested different hybrid optimization strategies.

We chose a naive multi-omics integration approach, such as early concatenation. While the predictive performances are encouraging, as discussed in [7], mixed integration strategies are often to be preferred and they could be tested to further improve the prediction power of a MINN-based method. Moreover, GPR rules could be leveraged to more directly link omics data to metabolic fluxes, for example by incorporating into future versions of MINN a loss term that maximizes the correlation between expression levels and predicted fluxes [2].

Additionally, more work is needed for assessing the interpretability of the flux distribution. As a first step in this direction, in our simulation we kept track of L_2 , as a proxy for how much the predicted metabolic profile complies with the theoretical assumptions of FBA. Moreover, this novel framework has been tested only for *E. coli*, the classical “work-horse” of microbial physiology. We hope the promising result of these works will prompt the creation of suitable datasets to apply these techniques to other microorganisms and to more diverse scenarios, in which secondary metabolism plays a bigger role and the information provided by the GEM is potentially even more effective in complementing the data-driven learning. In addition, the mechanistic component of MINN holds strong potential to improve predictions in more complex systems, such as eukaryotic cells or microbial communities, and the architecture of MINN is designed to scale seamlessly to these settings.

Although this was not the most favorable scenario for FBA, the results of the MINN-reservoir highlight its potential as a promising strategy to enable the full integration of FBA into a machine learning framework, effectively combining the advantages of both mechanistic and data-driven approaches.

Finally, with this work we provide a practical guide for choosing the most suitable MINN configuration based on the modeling objective. If the goal is only to achieve high predictive performance, the standard MINN configuration is sufficient. When the aim is to improve the quality of the predicted flux distribution, the optimization strategies presented here can reduce the mechanistic constraints violation. Lastly, if the objective is to enrich mechanistic models with additional inputs, the

MINN-reservoir offers a viable solution to generate constraints in a data-driven way, while keeping the structure and interpretability of classical FBA.

CRediT authorship contribution statement

Gabriele Tazza: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Francesco Moro:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Dario Ruggeri:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Bas Teusink:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **László Vidács:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Code availability

The code used for the analyses presented in this work is available at <https://github.com/gabrieletaz/MINN>.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this paper. Any opinions, findings, conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the funding bodies. Therefore, the authors declare that they have no conflicts of interest.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.csbj.2025.08.004>.

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